LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety precautions while working on Embedded System
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety precautions while working on Embedded System
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of ESD wrist strap usage and static-free workspace setup; Instructor supervision of tool handling and grounding verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is an Embedded System?
(b) Why is static electricity dangerous to electronics?
(c) List three general safety rules for electrical work.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Electrostatic Discharge (ESD) hazards • ESD protection methods and wrist straps • Safe handling of sensitive components • Workspace grounding requirements • Proper tool usage and storage	L & Ds	PPT, IB	15 min
(b)	• Power supply safety and isolation • Thermal management and ventilation • Emergency shutdown procedures • Cable management and strain relief • Personal protective equipment (PPE) usage	L & Ds	PPT, PDF	20 min
(c)	• Software safety and data integrity • Backup procedures before maintenance • Secure coding practices • Version control management • Incident reporting protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the primary cause of damage to embedded systems during handling.
(b) Describe the correct procedure for grounding yourself before touching a circuit board.

12. Home Assignment and Follow Up	Prepare a brief report on the specific safety protocols required for the Altimeter system discussed in LP 09, highlighting any unique ESD risks compared to general embedded systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and function of Altimeter and their types. Global positioning system (GPS) introduction, purpose airborne and space components of the system. Safety precautions to be observed whilst working on avionics systems	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Specific model of ESD wrist strap required for this course.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min